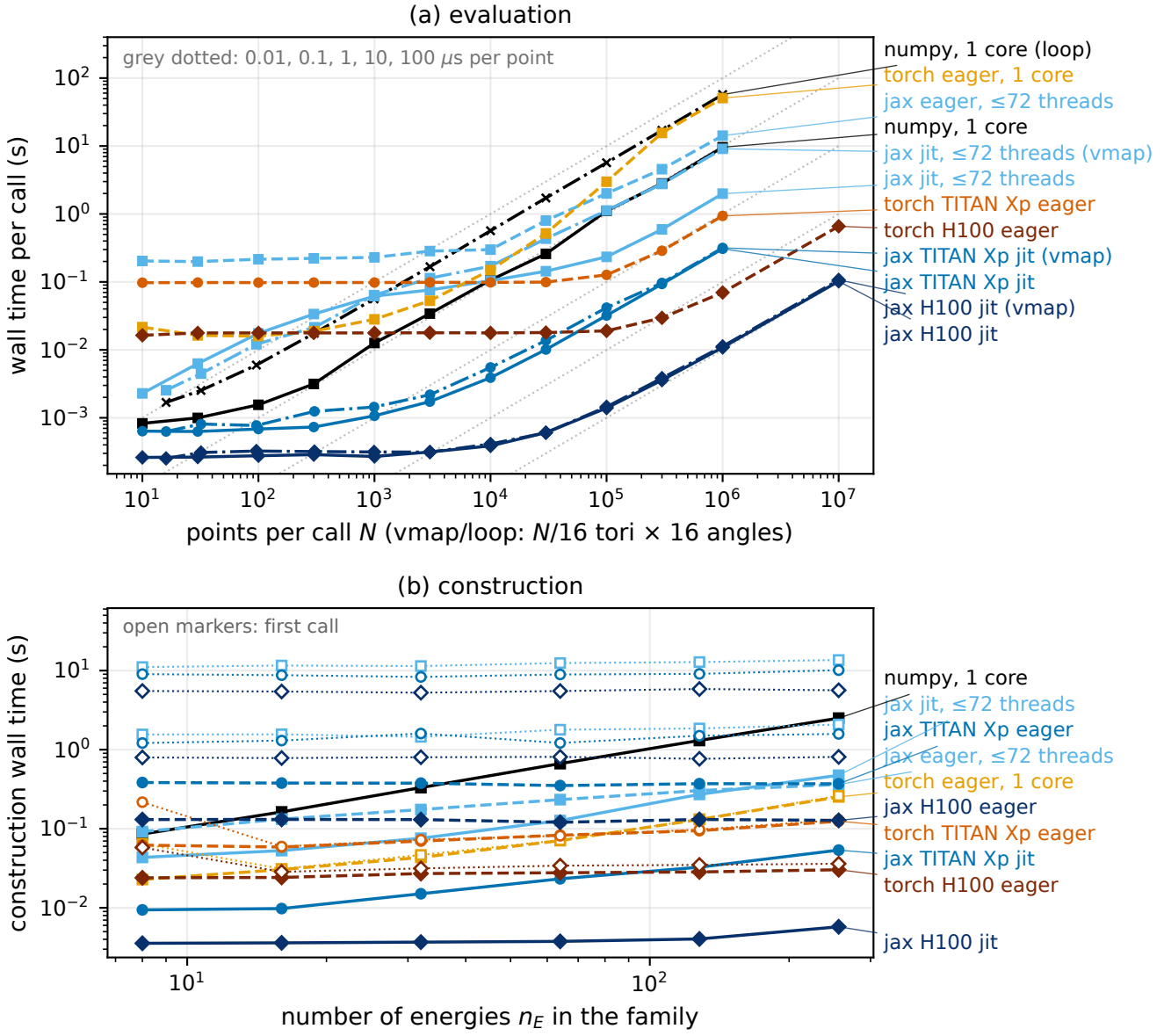
(a) evaluation
grey dotted: 0.01, 0.1, 1, 10, 100 μs per point
wall time per call (s)
points per call N (vmap/loop: N/16 tori × 16 angles)
numpy, 1 core (loop)
torch eager, 1 core
jax eager, ≤72 threads
numpy, 1 core
jax jit, ≤72 threads (vmap)
jax jit, ≤72 threads
torch TITAN Xp eager
torch H100 eager
jax TITAN Xp jit (vmap)
jax TITAN Xp jit
jax H100 jit (vmap)
jax H100 jit
(b) construction
open markers: first call
construction wall time (s)
number of energies n_E in the family
numpy, 1 core
jax jit, ≤72 threads
jax TITAN Xp eager
jax eager, ≤72 threads
torch eager, 1 core
jax H100 eager
torch TITAN Xp eager
jax TITAN Xp jit
torch H100 eager
jax H100 jit